

1675

2,920
MILESTHE GAP BETWEEN RINGS A AND
B OF **SATURN**, KNOWN AS THE
CASSINI DIVISION

ISAAC NEWTON PUBLISHED HIS HYPOTHESIS OF LIGHT in 1675, suggesting that light was made up of particles that he called corpuscles. Physicists had long debated the nature of light: some, like Newton, favored particles, others the theory that light traveled like waves. The corpuscular theory prevailed until the 1800s, when British physicist Thomas Young proved that light was wavelike (see 1801).

In March, Charles II appointed British astronomer **John Flamsteed** (1646–1719) as the first Astronomer Royal for a new observatory at Greenwich, London. The **Royal Observatory**

Royal Greenwich Observatory
Home of the prime meridian and Greenwich Mean Time, the Royal Observatory was made a World Heritage Site by UNESCO in 1997.

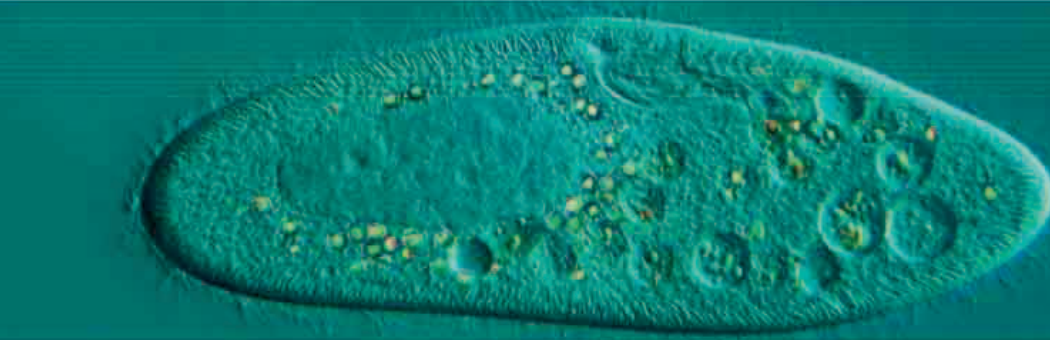


was built to improve ways of measuring longitude for sea navigation: it marked what later became the prime meridian between east and west. Many years later, through international agreement, it would mark an official starting point of each day—at the stroke of midnight **Greenwich Mean Time (GMT)**.

Italian microscopist **Marcello Malpighi** (1628–94) published his principal work *Anatome Plantarum* (*Anatomy of Plants*) on the fine structure of plant tissues, naming the outer layer of a leaf the epidermis and its tiny breathing pores, stomata. Another Italian, astronomer **Giovanni Cassini**, noticed that Saturn's distinctive ring was divided. The dark gap became known as the **Cassini division**.

Scientists now know that this gap comprises small particles at low density.

1676–78



Leeuwenhoek's animalcules were really single-celled organisms—such as this *Paramecium*—many of which reproduce rapidly in standing water.

IN 1676, DUTCH ASTRONOMER OLE RÖMER (1644–1710) used astronomical measurements to deduce that **light has a fixed speed**—something that was not readily accepted until the mid-1700s.

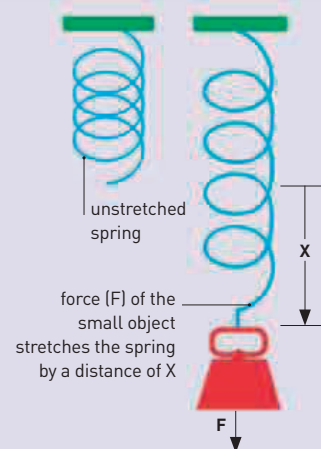
In 1668, Dutch textile merchant **Antony van Leeuwenhoek** (1632–1723) had traveled to London and been impressed by Englishman Robert Hooke's publication on microscopic life: **Micrographia**. On his return home, Leeuwenhoek designed his own **microscopes**—with small spherical lenses fashioned by drawing out a thread of glass, which was rounded off at the tip (see p.114). The magnifications rivaled those of any microscopes then in use—and he set about exploring miniature worlds. When he saw the microscopic taste buds on an ox's tongue, he was curious to study taste. This led him to soak pepper and spices in water. One of his pepper infusions ended up teeming with tiny living beings, which Leeuwenhoek referred to as **animalcules**. Many of these organisms were likely to have been the microbes that were later referred to as protozoans.

In 1676, Leeuwenhoek wrote the first of many letters to the Royal Society describing what he saw—initially provoking scepticism. The following year,

HOOKE'S LAW

Hooke originally applied his law of elasticity to a clock spring, but it applies to any elastic material—a solid that can change shape but then return to its original form.

As applied force (F) increases, so does the stretch length (X): doubling the force stretches the elastic twice as much. The law applies up to a certain elastic limit, beyond which the material does not recover and may snap.



he became the first person to see **human spermatozoa** and, as he persisted, Leeuwenhoek's scientific reputation improved.

In 1677, English astronomer **Edmond Halley** suggested it



Antony van Leeuwenhoek

This merchant's experience of using a magnifying glass in the textile trade led him to make microscopes, and then microbiological discoveries.

was possible to calculate the **distance between Earth and the Sun**—later known as the astronomical unit—by making geometric measurements during a **transit of Venus** as the planet passed in front of the Sun. Halley could not test this theory in his lifetime, but at the next transit, in 1761, his technique was used to produce a value very close to modern estimates.

In England, **Robert Hooke** had been turning his attention to the physics of elastic **clock springs**. He formalized an everyday observation—that the force applied is proportional to the amount of stretch—into what became known as **Hooke's Law**, which he published in 1678.

March John Flamsteed is appointed Britain's Astronomer Royal

Isaac Newton proposes that light is made up of particles

1676 Ole Rømer calculates the speed of light

November 1677 Leeuwenhoek sees spermatozoa in the microscope

1678 Edmond Halley publishes first catalog of stars seen in the southern sky

Marcello Malpighi publishes his *Anatomy of Plants*

Giovanni Cassini discovers a gap in Saturn's rings

October 9, 1676 Antony van Leeuwenhoek describes animalcules

October 5, 1677 Robert Hooke confirms Leeuwenhoek's microbe observations

1678 Hooke describes the elastic properties of a spring, in what becomes known as Hooke's Law